

“**Sawfly**（锯蜂）”是一种与黄蜂相似的昆虫，这种昆虫会切开植物并把自己的卵产在切口处。苏格兰赫瑞-瓦特大学的研究人员一直在研究雌性锯蜂怎么知道如何做这种独特的切割，从而避免杀死或损伤植物。

Sawflies are named after the saw-like **organ** used by the females to **lay** eggs in plants. The insects want to avoid killing the plants, which provide food for their **larvae**.

锯蜂以它身上一种像锯子一样的器官命名，雌性锯蜂会用这个锯状器官在植物内部产卵。这种昆虫会尽量避免杀死植物，因为植物可以为它们的幼虫提供食物。

The researchers discovered that small **serrations** on the sawfly's teeth worked with larger **protrusions** to create a selective cutting action, allowing them to avoid cutting internal structures carrying water and nutrients. The team **scaled up** the **mechanism** and tested it on material **mimicking** human tissue.

研究人员发现，锯蜂牙齿上较小的锯齿状突起和较大的突起物共同作用，从而创造出了一种具有选择性的切割动作，这种动作让锯蜂能够避免切断植物中含有水分和养分的内部结构。研究团队按比例放大了这一构造机制，然后在模拟了人体组织的材料上对其进行了测试。

Although more work is needed, they think there's potential for a **surgical** instrument based on this natural mechanism, which could **instinctively** avoid critical **tissues** whilst cutting.

虽然尚有更多工作需要完成，但研究人员认为也许能够发明出一种基于这种自然机制的外科手术器械，这种器械可以在切割时本能地避开关键组织。

词汇表

organ	器官
lay	产（卵）
larvae	幼虫
serrations	锯齿状
protrusions	突起物，凸出物
scale up	等比例放大
mechanism	构造，机制
mimicking	模拟，模仿
surgical	外科手术用的
instinctively	天然地，本能地
tissues	（生物的）组织

听力练习（答案见下页）

一、听第一遍音频，回答问题。

1. What are researchers hoping to use this information to develop?
 - a. Insect-like robots for use in surgery.
 - b. Instruments for surgeons that cause less damage to patients.

二、用一分钟阅读下列问题，然后听第二遍音频并回答问题。

1. Sawflies use their saw-like organ to get food.
 - a. True
 - b. False
 - c. Not given
2. Different shaped parts of the sawfly's teeth allow them to cut _____.
 - a. at different speeds
 - b. only what they need to
 - c. tough plant fibres
3. Researchers tested a larger version of the sawfly's organ on...
 - a. samples from surgical patients.
 - b. something similar to human flesh.
 - c. internal structures of plants.
4. What do researchers hope to develop?
 - a. something that only cuts certain tissues
 - b. something that cuts automatically
 - c. something that only cuts diseased tissue

三、听第三遍音频，将句子补充完整。（提示：可以反复听，直到听清楚每一个词）

1. The insects _____ killing the plants, which provide food for their larvae.
2. The team scaled up the mechanism and _____ material mimicking human tissue.

答案

一、听第一遍音频，回答问题。

1. What are researchers hoping to use this information to develop?

Instruments for surgeons that cause less damage to patients

二、用一分钟阅读下列问题，然后听第二遍音频并回答问题。

1. **b. false.** Sawflies are named after the saw-like organ used by the females **to lay eggs in plants.**
2. **b.** The researchers discovered that small serrations on the sawfly's teeth worked with larger protrusions to create a **selective** cutting action, allowing them to **avoid cutting internal structures carrying water and nutrients.**
3. **b.** The team scaled up the mechanism and tested it on material **mimicking human tissue.**
4. **a.** Although more work is needed, they think there's potential for a surgical instrument based on this natural mechanism, which could **instinctively avoid critical tissues whilst cutting.**

三、听第三遍音频，将句子补充完整。（提示：可以反复听，直到听清楚每一个词）

1. The insects **want to avoid** killing the plants, which provide food for their larvae.
2. The team scaled up the mechanism and **tested it on** material mimicking human tissue.